

GVPT622 Lab Session

Descriptive Statistics in R

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University of Maryland

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- ▶ **tidyverse**, **janitor**, **skimr**. Install them now if you have not:

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install.packages(c("tidyverse", "janitor", "skimr"))
```

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```
install.packages(c("tidyverse", "janitor", "skimr"))
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- ▶ Please don't hesitate to ask me questions when something is not running

Set Up



- ▶ A roll call vote in Congress is a procedure where each legislator's individual vote ("yea," "nay," or "present") is documented

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- ▶ [voteview.com](https://www.voteview.com) has recorded every vote dating back to the 1st Congress (1789)
- ▶ What is present in their data is also when a legislator does not vote
- ▶ I calculated a roll call attendance rate through a simple equation:

$$\text{Attendance Rate} = \frac{\text{Roll Call Votes Observed}}{\text{Total Roll Calls in Congress}} \quad (1)$$

- ▶ A roll call vote in Congress is a procedure where each legislator's individual vote ("yea," "nay," or "present") is documented
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- ▶ What is present in their data is also when a legislator does not vote
- ▶ I calculated a roll call attendance rate through a simple equation:

$$\text{Attendance Rate} = \frac{\text{Roll Call Votes Observed}}{\text{Total Roll Calls in Congress}} \quad (1)$$

- ▶ I then merged data that we used before, courtesy of the [Center for Effective Lawmaking](#)

Building the Dataset

```
library(tidyverse)

attendance <- read_csv("attendance_cel_panel.csv") |>
  filter(chamber == "House", source == "both") |>
  transmute(
    member      = cel_name,
    congress    = congress,
    year        = cel_year,
    state       = state_abbrev,

    # four forms of the same quantity: showing up to vote
    attendance   = 100 * attendance_rate,
    absence      = 100 - 100 * attendance_rate,
    missed_votes = votes_eligible - votes_attended,
    term_share   = 100 * share_of_chamber_rollcalls,
```

Building the Dataset (continued)

```
# covariates
party      = factor(ifelse(dem == 1, "Democrat", "Republican")),
majority   = factor(ifelse(majority == 1, "Majority", "Minority"),
                    levels = c("Minority", "Majority")),
role       = factor(ifelse(chair == 1, 3,
                            ifelse(subchr == 1, 2, 1)),
                    levels = c(1, 2, 3),
                    labels = c("Rank and file", "Subcmte chair",
                               "Cmte chair")),
freshman   = factor(ifelse(freshman == 1, "Freshman", "Returning"),
                    levels = c("Returning", "Freshman")),
female     = female,      vote_pct    = vote_pct,
ideology    = nominate_dim1, seniority = seniority,
deleg_size  = deleg_size,  party_size = party_size
)
```

Using head()

```
head(attendance)
```

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```

```
# A tibble: 6 x 18
  member      congress year state attendance absence missed_votes
  <chr>      <dbl> <dbl> <chr>      <dbl>      <dbl>      <dbl>
1 Abdnor, James      93  1973 SD          93.2        6.77         73
2 Abzug, Bella       93  1973 NY          96.5        3.53         38
3 Adams, Brock       93  1973 WA          89.8       10.2        110
4 Addabbo, Jos~     93  1973 NY          90.9        9.09         98
5 Albert, Carl      93  1973 OK         100          0           0
6 Alexander, B~     93  1973 AR          82.8       17.2        185
# i 11 more variables: term_share <dbl>, party <fct>,
#   majority <fct>, role <fct>, freshman <fct>, female <dbl>,
#   votepct <dbl>, ideology <dbl>, seniority <dbl>,
#   deleg_size <dbl>, party_size <dbl>
```

Describing Your Data

In Tidyverse: `skimr::skim()`

```
library(skimr)  
  
skim(attendance$attendance)
```

In Tidyverse: skimr::skim()

```
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```

```
skim(attendance$attendance)
```

```
-- Data Summary -----
```

	Values
Name	attendance\$attendance
Number of rows	11470

```
...
```

```
-- Variable type: numeric -----
```

	skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75
1	data	0	1	93.9	8.02	0	92.3	96.3	98.3
	p100								
1	100								

In Tidiverse: `skimr::skim()`

```
library(skimr)

skim(attendance$attendance)
```

```
-- Data Summary -----
Name                               Values
Number of rows                    11470
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-- Variable type: numeric -----
skim_variable n_missing complete_rate mean  sd p0  p25  p50  p75
1 data              0           1 93.9 8.02  0 92.3 96.3 98.3
  p100
1 100
```

- ▶ The same five-number summary as the `summary()` function, plus missing-data counts and descriptive statistics in one table

In Tidiverse: skimr::skim()

```
library(skimr)

skim(attendance$attendance)
```

```
-- Data Summary -----
Name                               Values
Number of rows                    11470
...
-- Variable type: numeric -----
skim_variable n_missing complete_rate mean  sd p0  p25  p50  p75
1 data              0             1 93.9 8.02 0 92.3 96.3 98.3
  p100
1 100
```

- ▶ The same five-number summary as the `summary()` function, plus missing-data counts and descriptive statistics in one table
- ▶ Always run this on any new variable

What R Thinks Each Variable Is

```
sapply(attendance[, c("member", "congress", "attendance",  
                      "missed_votes", "party", "role",  
                      "female")],  
       class)
```

What R Thinks Each Variable Is

```
sapply(attendance[, c("member", "congress", "attendance",  
                      "missed_votes", "party", "role",  
                      "female")],  
       class)
```

member	congress	attendance	missed_votes	party
"character"	"numeric"	"numeric"	"numeric"	"factor"
role	female			
"factor"	"numeric"			

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                      "missed_votes", "party", "role",  
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```

member	congress	attendance	missed_votes	party
"character"	"numeric"	"numeric"	"numeric"	"factor"
role	female			
"factor"	"numeric"			

► `class()` reports the type of variable, not the level of measurement

What R Thinks Each Variable Is

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sapply(attendance[, c("member", "congress", "attendance",  
                      "missed_votes", "party", "role",  
                      "female")],  
       class)
```

member	congress	attendance	missed_votes	party
"character"	"numeric"	"numeric"	"numeric"	"factor"
role	female			
"factor"	"numeric"			

- ▶ `class()` reports the type of variable, not the level of measurement
- ▶ `female` is stored as `numeric` but is dichotomous; `congress` is stored as `numeric` but is actually an identifier. R does not know that — you have to understand the codebook

- ▶ A nominal variable is a **name** attached to a **numeric code**. In R, that pairing is a **factor**

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Coding a Nominal Variable

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```
levels(attendance$party)  
as.integer(attendance$party)[1:15]
```

Coding a Nominal Variable

- ▶ A nominal variable is a **name** attached to a **numeric code**. In R, that pairing is a **factor**

```
levels(attendance$party)
as.integer(attendance$party)[1:15]
```

```
[1] "Democrat"    "Republican"
[1] 2 1 1 1 1 1 1 2 1 2 1 2 2 2 2
```

Coding a Nominal Variable

- ▶ A nominal variable is a **name** attached to a **numeric code**. In R, that pairing is a **factor**

```
levels(attendance$party)
as.integer(attendance$party)[1:15]
```

```
[1] "Democrat"    "Republican"
[1] 2 1 1 1 1 1 1 2 1 2 1 2 2 2 2
```

- ▶ Democrat → 1, Republican → 2. The codes are labels, **not** quantities

Dichotomous Variables

```
table(attendance$female)  
mean(attendance$female)
```

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```
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```

```
      0      1
10029  1441
[1] 0.1256321
```

Dichotomous Variables

```
table(attendance$female)
mean(attendance$female)
```

```
      0      1
10029 1441
[1] 0.1256321
```

- ▶ Coded 0/1, a dichotomous variable has a useful property: its **mean is a proportion**

Dichotomous Variables

```
table(attendance$female)
mean(attendance$female)
```

```
      0      1
10029  1441
[1] 0.1256321
```

- ▶ Coded 0/1, a dichotomous variable has a useful property: its **mean is a proportion**
- ▶ 12.56 percent of member-Congresses in this panel are women

Dichotomous Variables

```
table(attendance$female)
mean(attendance$female)
```

```
      0      1
10029 1441
[1] 0.1256321
```

- ▶ Coded 0/1, a dichotomous variable has a useful property: its **mean is a proportion**
- ▶ 12.56 percent of member-Congresses in this panel are women
- ▶ This is a case where averaging a category code is meaningful — because the codes are 0 and 1

Collapsing in R: cut()

```
att_cat <- cut(attendance$attendance,  
              breaks = c(0, 90, 97, 100),  
              include.lowest = TRUE,  
              labels = c("Low", "Middle", "High"))  
  
table(att_cat)
```

Collapsing in R: cut()

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att_cat <- cut(attendance$attendance,  
              breaks = c(0, 90, 97, 100),  
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table(att_cat)
```

```
att_cat  
  Low Middle   High  
1920  4699  4851
```

Collapsing in R: cut()

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att_cat <- cut(attendance$attendance,  
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table(att_cat)
```

```
att_cat  
  Low Middle   High  
1920   4699   4851
```

- **breaks** establishes the cut points; **include.lowest = TRUE** keeps the members at exactly 0

Describing Categorical Variables

Frequency Distribution: `table()`

```
table(attendance$role)
```

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```
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```

Rank and file	Subcmte chair	Cmte chair
8337	2557	576

From Counts to Percentages

```
round(100 * prop.table(table(attendance$role)), 1)
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Rank and file	Subcmte chair	Cmte chair
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From Counts to Percentages

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round(100 * prop.table(table(attendance$role)), 1)
```

Rank and file	Subcmte chair	Cmte chair
72.7	22.3	5.0

- ▶ `prop.table()` divides every cell by the total; multiplying by 100 makes it a percentage
- ▶ Nearly three-quarters of member-Congress observations are rank-and-file. Only 5% of observations hold a full committee gavel

In Tidyverse: `janitor::tabyl()`

```
library(janitor)

tabyl(attendance, role)
```

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	role	n	percent
Rank and file		8337	0.72685266
Subcmte chair		2557	0.22292938
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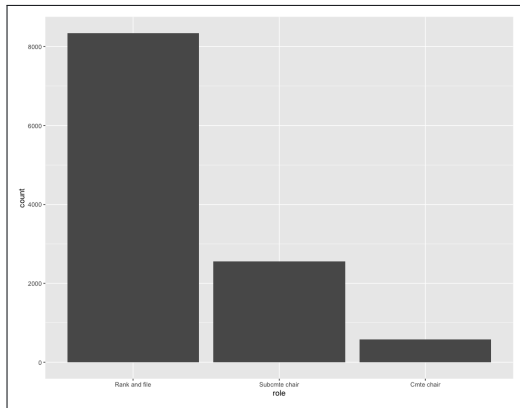
- ▶ One line of code gives counts **and** percentages together
- ▶ `tabyl()` adds a **percent** column too, once a variable actually has missing values to exclude from the denominator

Barplot in ggplot2

```
ggplot(attendance) +  
  aes(x = role) +  
  geom_bar()
```

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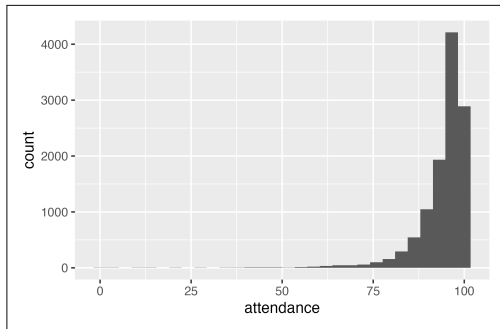
Describing Continuous Variables

Histogram in ggplot2

```
ggplot(attendance) +  
  aes(x = attendance) +  
  geom_histogram(bins = 30)
```

Histogram in ggplot2

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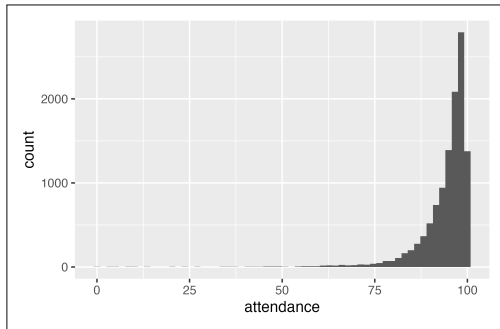


The Number of Bins is up to You

```
ggplot(attendance) +  
  aes(x = attendance) +  
  geom_histogram(bins = 60)
```

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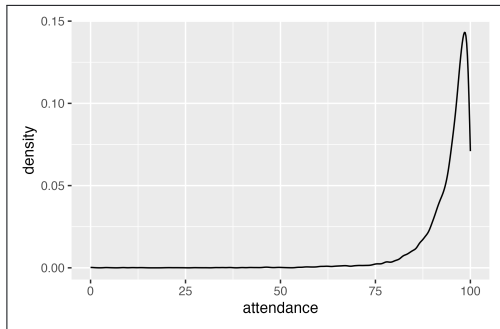


Density Plot in ggplot2

```
ggplot(attendance) +  
  aes(x = attendance) +  
  geom_density()
```

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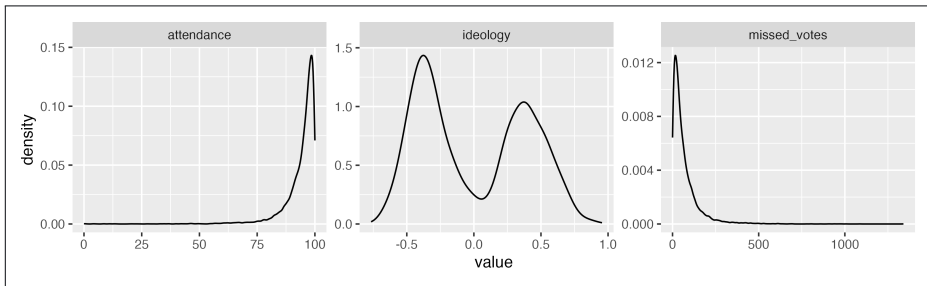


Three Shapes

```
shape_dat <- bind_rows(  
  tibble(value = attendance$attendance, panel = "attendance"),  
  tibble(value = attendance$missed_votes, panel = "missed_votes"),  
  tibble(value = na.omit(attendance$ideology), panel = "ideology"))  
  
ggplot(shape_dat) + aes(x = value) + geom_density() +  
  facet_wrap(~ panel, scales = "free", nrow = 1)
```

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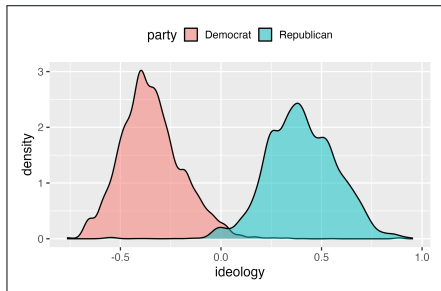


How Many Modes?

```
ggplot(filter(attendance, !is.na(ideology))) +  
  aes(x = ideology, fill = party) +  
  geom_density(alpha = 0.5)
```

How Many Modes?

```
ggplot(filter(attendance, !is.na(ideology))) +  
  aes(x = ideology, fill = party) +  
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```



Measures of Central Tendency

The Mean

```
mean(attendance$attendance)
```

The Mean

```
mean(attendance$attendance)
```

```
[1] 93.89901
```

The Mean

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mean(attendance$attendance)
```

```
[1] 93.89901
```

- ▶ The average House member votes on 93.89 percent of roll calls

The Mean

```
mean(attendance$attendance)
```

```
[1] 93.89901
```

- ▶ The average House member votes on 93.89 percent of roll calls
- ▶ The mean uses every observation — which is exactly why one member at 0 percent moves it

The Median

```
median(attendance$attendance)
```

The Median

```
median(attendance$attendance)
```

```
[1] 96.2726
```

The Median

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```

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[1] 96.2726
```

- ▶ Half of all member-Congresses fall above 96.27 percent

The Median

```
median(attendance$attendance)
```

```
[1] 96.2726
```

- ▶ Half of all member-Congresses fall above 96.27 percent
- ▶ Median $>$ mean is the signature of a **left-skewed** distribution

- ▶ R has no built-in mode function — `mode()` reports the storage type, not the most common value

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head(sort(table(attendance$attendance),  
         decreasing = TRUE), 5)
```

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- ▶ Tabulate and sort instead:

```
head(sort(table(attendance$attendance),  
         decreasing = TRUE), 5)
```

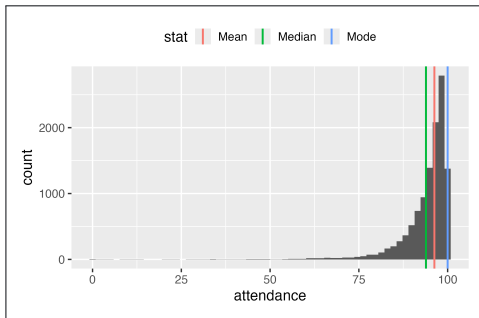
100	99.3976	99.498	99.5984	99.7992
182	23	22	22	22

Graphical Display of Central Tendency

```
ggplot(attendance) +  
  aes(x = attendance) +  
  geom_histogram(bins = 60) +  
  geom_vline(xintercept = c(100, 96.27, 93.89))
```

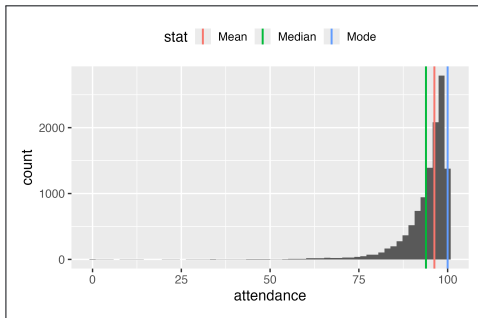
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```



- ▶ Three representations of member attendance; whichever you report should be defensible

Five-Number Summary, Box Plots, and Outliers

The Five-Number Summary

- ▶ Minimum, Q_1 , median, Q_3 , maximum — in that order

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```
fivenum(attendance$attendance)
```

The Five-Number Summary

- ▶ Minimum, Q_1 , median, Q_3 , maximum — in that order

```
fivenum(attendance$attendance)
```

```
[1] 0.0000 92.3333 96.2726 98.2847 100.0000
```

The Five-Number Summary

- ▶ Minimum, Q_1 , median, Q_3 , maximum — in that order

```
fivenum(attendance$attendance)
```

```
[1] 0.0000 92.3333 96.2726 98.2847 100.0000
```

- ▶ Five numbers, and you already know the median, the spread of the middle half, and both extremes

summary() on Attendance

```
summary(attendance$attendance)
```

summary() on Attendance

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summary(attendance$attendance)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	92.33	96.27	93.90	98.28	100.00

summary() on Attendance

```
summary(attendance$attendance)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	92.33	96.27	93.90	98.28	100.00

- Six numbers, one line: Minimum, Q_1 , median, mean, Q_3 , maximum

```
summary(attendance$attendance)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	92.33	96.27	93.90	98.28	100.00

- ▶ Six numbers, one line: Minimum, Q_1 , median, mean, Q_3 , maximum
- ▶ The five-number summary and the mean, in a single call

Interquartile Range

```
IQR(attendance$attendance)
```

Interquartile Range

```
IQR(attendance$attendance)
```

```
[1] 5.9503
```

Interquartile Range

```
IQR(attendance$attendance)
```

```
[1] 5.9503
```

► Or by hand, which is all `IQR()` does:

Interquartile Range

```
IQR(attendance$attendance)
```

```
[1] 5.9503
```

► Or by hand, which is all `IQR()` does:

Interquartile Range

```
IQR(attendance$attendance)
```

```
[1] 5.9503
```

► Or by hand, which is all `IQR()` does:

```
quantile(attendance$attendance, 0.75) -  
  quantile(attendance$attendance, 0.25)
```

Interquartile Range

```
IQR(attendance$attendance)
```

```
[1] 5.9503
```

► Or by hand, which is all `IQR()` does:

```
quantile(attendance$attendance, 0.75) -  
  quantile(attendance$attendance, 0.25)
```

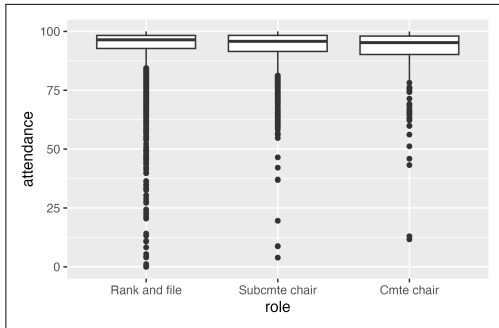
```
75%  
5.9503
```

Boxplot in ggplot2

```
ggplot(attendance) +  
  aes(x = role, y = attendance) +  
  geom_boxplot()
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The $1.5 \times \text{IQR}$ Rule, by Hand

```
q1 <- quantile(attendance$attendance, 0.25)
q3 <- quantile(attendance$attendance, 0.75)
iqr <- q3 - q1

round(c(lower_fence = q1 - 1.5 * iqr,
        upper_fence = q3 + 1.5 * iqr), 2)
```

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```
lower_fence upper_fence
      83.41      107.21
```

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```

```
lower_fence upper_fence
      83.41      107.21
```

- The upper fence is 107.21%, which is impossible. A bounded variable can have a fence outside its own range

Measures of Spread

Minimum and Maximum

```
min(attendance$attendance)  
max(attendance$attendance)
```

Minimum and Maximum

```
min(attendance$attendance)  
max(attendance$attendance)
```

```
[1] 0  
[1] 100
```

Minimum and Maximum

```
min(attendance$attendance)  
max(attendance$attendance)
```

```
[1] 0  
[1] 100
```

- ▶ A member with 0 percent attendance is real — a few members were sworn in but died before casting a vote

The Range

```
max(attendance$attendance) - min(attendance$attendance)
```

The Range

```
max(attendance$attendance) - min(attendance$attendance)
```

```
[1] 100
```

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► If you need it often, write it once:

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► If you need it often, write it once:

```
range2 <- function(x) {  
  max(x) - min(x)  
}  
  
range2(attendance$missed_votes)
```

The Range

```
max(attendance$attendance) - min(attendance$attendance)
```

```
[1] 100
```

► If you need it often, write it once:

```
range2 <- function(x) {  
  max(x) - min(x)  
}  
  
range2(attendance$missed_votes)
```

```
[1] 1340
```

Standard Deviation and Variance

```
sd(attendance$attendance)  
var(attendance$attendance)
```

Standard Deviation and Variance

```
sd(attendance$attendance)  
var(attendance$attendance)
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```
[1] 8.019767  
[1] 64.31667
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Standard Deviation and Variance

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- ▶ The variance is the standard deviation squared

Standard Deviation and Variance

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sd(attendance$attendance)
var(attendance$attendance)
```

```
[1] 8.019767
[1] 64.31667
```

- ▶ The variance is the standard deviation squared
- ▶ The standard deviation is in the units of the variable (percentage points), so it is the one you report

Wrap-Up

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- ▶ Next week: Research Design

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- ▶ Report the mean and median of `seniority`.
- ▶ Report the range, variance, and standard deviation of `votepct`

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Questions?